

Rauzy Path Decoding and an Infinite-Fibre Noncompactness Obstruction for an AGY Induced Transfer Operator

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Abstract

We prove that the unsmoothed oscillator twist of an Avila–Gouëzel–Yoccoz induced countable Rauzy transfer operator is noncompact on two natural spaces. The scalar construction is quasicompact on bounded C^1 functions, but adjoining the infinite-dimensional metaplectic fibre creates an exact obstruction: a branch-supported C^1 bump followed by point evaluation compresses the full operator to a nonzero scalar multiple of one oscillator unitary. This proves noncompactness throughout the AGY half-plane $\Re s > -\sigma_0$. Put $\mathcal{F} = L^2(\mathbb{R}^2)$. After invariant-density normalization, the operator is contractive but noncompact on $L^2(\mu; \mathcal{F})$ for $\Re s \geq 0$; on $s = it$ it is a coisometry with essential norm one. We note that this normalized L^2 obstruction already occurs for the scalar Perron–Frobenius operator and is therefore a space-level robustness test, not an oscillator-specific novelty. We also give an all-length row-subtraction decoder showing that a Rauzy path from a fixed labeled permutation is uniquely determined by its full chronological matrix. Thus distinct return branches cannot cancel through equal matrices and opposite metaplectic signs in the four-letter full-rank model. A project-chosen deterministic length-128 word with eight complete blocks fixes one neat strongly-positive AGY-admissible section, and an independent implementation replays its exact matrix and 35,420 finite decoder stress cases. The result rules out ordinary nuclear Fredholm determinants on the two stated spaces; holomorphic no-localizer spaces and generalized traces remain separate open candidates.

1 Introduction

A transfer-operator proposal for Hilbert–Pólya must eventually choose an operator space. Formal periodic weights do not decide whether the proposed operator is compact, nuclear, or even bounded, and scalar spectral estimates need not survive an infinite-dimensional quantum fibre. This distinction is decisive for a Rauzy–Veech proposal coupled to the oscillator representation of its symplectic homology cocycle.

The predecessor calculation reconstructed a four-letter genus-two Rauzy class and proved abstract noncompactness criteria for discrete metaplectic extensions. Its unresolved hypothesis was concrete: no published transfer space had yet been shown to satisfy the branch-compression or norm-summability hypotheses. We verify it here using the precompact first-return model of Avila, Gouëzel, and Yoccoz (AGY) [2]. AGY constructs a countable full-branch expanding map, a roof with exponential tails, and a weighted operator on bounded C^1 functions. Once an inducing word is fixed, this is an actual analytic operator family rather than an averaged Rauzy graph.

Our one-sentence conclusion is the following. For every AGY neat-return model satisfying the source hypotheses below, adjoining the unsmoothed oscillator fibre makes the raw vector-valued C_b^1 operator noncompact throughout its defining half-plane; the normalized L^2 companion is already noncompact at scalar level and remains so after the twist throughout $\Re s \geq 0$; full-matrix Rauzy decoding also removes the proposed same-matrix branch-cancellation escape.

The argument is elementary once the correct space and chronology are fixed, but it is stronger than finding a noncompact formal summand. On C_b^1 , the branch series converges absolutely in operator norm. We then localize the *full* sum with a source-provided bump and a bounded evaluation map. All other branches vanish in the compression. On the normalized Hilbert space, a cylinder projection gives a second exact isolation, while a twisted Koopman operator identifies the imaginary-axis transfer as a coisometry.

The paper makes four checkable contributions.

1. We freeze a project-chosen four-letter AGY-admissible section with $\gamma_* = t^{64}(tbttbtbb)^8$. Eight complete blocks prove strong positivity by the AGY $3d - 4$ criterion; a maximal-prefix and no-border check proves neatness.
2. We prove an all-length decoder. From a fixed labeled start, the true first Rauzy arrow is the unique one whose winner row dominates its loser row in B_γ^T . Row subtraction recovers the remaining path matrix and strictly decreases its entry sum.
3. We prove absolute branch-norm convergence and an exact bump/evaluation compression on $C_b^1(\Delta; L^2(\mathbb{R}^2))$. The compressed operator is a nonzero infinite-dimensional unitary, so the full operator is noncompact and not Banach nuclear for every $\Re s > -\sigma_0$.
4. We prove that the invariant-density-normalized operator on $L^2(\mu; L^2(\mathbb{R}^2))$ is contractive but noncompact for $\Re s \geq 0$. For $s = it$, its essential norm is exactly one. In the farther-right half-plane $\Re s \geq d/2$, the explicit Hilbert branch atoms are absolutely summable in operator norm.

The result is deliberately space-specific. Noncompactness removes an ordinary nuclear or Schatten-class determinant on the tested realizations; it does not say that $I - z\mathcal{L}_s$ is never Fredholm. A holomorphic space has no nonzero branch-supported bump, and a flat, distributional, or semifinite trace need not arise from an ordinary compact operator. These options are not repairs of the present operator and require independent source specifications.

Relation to existing transfer spaces. AGY's scalar C^1 space is the main source object. The quasi-Hölder realization of Aimino, Nicol, and Todd [1] confirms that other regularity completions are useful for Rauzy–Veech–Zorich statistics, but it does not state that every return-cylinder indicator belongs to its space. We therefore do not use quasi-Hölder cylinder projectors. Compactness obstructions for a single operator-weighted composition map are already standard: the point-evaluation factorization of Bonet et al. [3] forces compact point weights. In a dynamical setting, Magee and Naud [6] allow arbitrary Hilbert-space unitary twists for norm estimates but establish the ordinary Fredholm determinant only for finite-dimensional twists. Our contribution is consequently not the bare observation that an infinite-dimensional unitary is noncompact. It is the exact isolation of that unitary from the convergent *multi-branch* AGY operator, with the stated full half-planes and the independent normalized L^2 conclusion. The oscillator representation and its metaplectic covariance are themselves standard [4].

Organization. Section 2 fixes the AGY model and the deterministic section. Section 3 proves matrix decoding. Sections 4 and 5 prove the two operator obstructions. Section 6 records exact independent checks, and Section 7 states the Route-A decision and surviving scope.

2 The source-specified AGY induced return model

2.1 Rauzy chronology

Let $\mathcal{A} = \{1, 2, 3, 4\}$. A labeled permutation is displayed as two ordered rows. At every irreducible state, the rightmost top and bottom labels are distinct. A top arrow has the top label

as winner and the bottom label as loser; a bottom arrow reverses those roles [7]. Following AGY, an arrow with winner w and loser ℓ carries

$$B_e = I + E_{\ell, w}. \quad (1)$$

If $\gamma = e_1 \cdots e_n$ is chronological, then later arrows multiply on the left:

$$B_\gamma = B_{e_n} \cdots B_{e_1}. \quad (2)$$

This convention is also the natural length-visitation convention after transposition. No frequency or transition average appears.

For a state π , let Ω_π be its crossing form in fixed label order. Every open edge transports the changing form,

$$B_e \Omega_{\text{src}} B_e^T = \Omega_{\text{dst}}. \quad (3)$$

A closed path at π therefore satisfies $B_\gamma^T J_\pi B_\gamma = J_\pi$, where $J_\pi = \Omega_\pi^{-1}$. In the four-letter reversal class all crossing forms have determinant one, so relative and absolute homology have the same four-dimensional rank.

To make the pathwise lift precise, use the base-state form $J_0 = J_{\pi_*}$ as a fixed reference and, for each Rauzy state π , choose an integral frame $S_\pi \in \text{GL}(4, \mathbb{Z})$ satisfying

$$S_\pi^T J_\pi S_\pi = J_0.$$

Such frames exist because the forms are integral, alternating, and unimodular. For an edge $e : \pi_{\text{src}} \rightarrow \pi_{\text{dst}}$, put

$$g_e = S_{\pi_{\text{dst}}}^{-1} B_e S_{\pi_{\text{src}}}. \quad (4)$$

Inverting (3) gives $B_e^T J_{\text{dst}} B_e = J_{\text{src}}$, hence $g_e^T J_0 g_e = J_0$. Moreover the intermediate frames telescope:

$$g_\gamma = g_{e_n} \cdots g_{e_1} = S_{\pi_{\text{end}}}^{-1} B_\gamma S_{\pi_{\text{start}}}.$$

Thus every g_e lies in $\text{Sp}(J_0, \mathbb{Z}) = \{g \in \text{GL}(4, \mathbb{Z}) : g^T J_0 g = J_0\}$, which is integrally conjugate to the standard $\text{Sp}(4, \mathbb{Z})$. After one fixed real identification of $(\mathbb{R}^4, J_0)$ with standard phase space, we choose a lift of every g_e and compose those lifts chronologically in the corresponding metaplectic group. Their oscillator images

$$U_\gamma = \mu_{\text{osc}}(\tilde{g}_\gamma) \quad \text{on} \quad \mathcal{F} = L^2(\mathbb{R}^2) \quad (5)$$

are unitary. The central sign is retained.

2.2 The AGY induced map

AGY chooses a strongly-positive neat loop γ_* at a Rauzy state and uses it to define a precompact first-return section [2]. The base is a bounded John domain Δ , equipped with its induced Lebesgue measure m , and partitioned modulo measure zero into countably many open sets D_γ . In particular m , and hence every measure with a positive density relative to m , is nonatomic. Each forward branch $T : D_\gamma \rightarrow \Delta$ is a C^1 diffeomorphism. We write $h_\gamma : \Delta \rightarrow D_\gamma$ for the inverse branch, $r_\gamma = r \circ h_\gamma$ for the branch roof, and $j_\gamma = \text{Jac } h_\gamma$ for the inverse Jacobian.

The source estimates provide constants independent of γ such that

$$\|Dh_\gamma\|_\infty \leq \kappa < 1, \quad \|D \log j_\gamma\|_\infty \leq C_J, \quad \|Dr_\gamma\|_\infty \leq C_r. \quad (6)$$

For $d = |\mathcal{A}|$, the projective formula is

$$h_\gamma(x) = \frac{B_\gamma^T x}{\|B_\gamma^T x\|_1}, \quad r_\gamma(x) = \log \|B_\gamma^T x\|_1, \quad j_\gamma(x) = e^{-dr_\gamma(x)}. \quad (7)$$

For the AGY return branches, $r_\gamma \geq \log 2 > 0$. The roof has exponential tails: for some $\sigma_0 > 0$,

$$\int_{\Delta} e^{\sigma_0 r} dm < \infty. \quad (8)$$

The map preserves a probability measure $d\mu = \rho dm$, where

$$0 < c_\rho \leq \rho \leq C_\rho < \infty, \quad \rho \in C_b^1(\Delta). \quad (9)$$

The scalar weighted operator used by AGY is

$$(L_s u)(x) = \sum_{\gamma} e^{-sr_\gamma(x)} j_\gamma(x) u(h_\gamma x), \quad \Re s > -\sigma_0, \quad (10)$$

on $C_b^1(\Delta)$. The metaplectic extension below keeps the same base weight and inserts U_γ without smoothing.

2.3 A deterministic four-letter section

The explicit reversal seed $(1234)/(4321)$ generates seven labeled states. We choose the AGY-admissible section base

$$\pi_* = (1342)/(4321).$$

At this state, set

$$\eta = tbttbttbb, \quad \gamma_* = t^{64}\eta^8. \quad (11)$$

The loop η is complete: its winners contain all four labels. The decomposition $(t^{64}\eta)\eta^7$ therefore consists of eight complete paths. Since $3d - 4 = 8$, the AGY criterion proves strong positivity. The maximal initial top run has length 65, at least half of $|\gamma_*| = 128$, and the final arrow is bottom. This is AGY's sufficient shape for neatness. An exact scan also finds no nonempty proper word that is both a prefix and suffix.

The chronological matrix is

$$B_{\gamma_*} = \begin{pmatrix} 18540 & 1210580 & 11430 & 27373 \\ 24020 & 1568410 & 14783 & 35450 \\ 50233 & 3279928 & 31130 & 74253 \\ 38803 & 2533625 & 24020 & 57343 \end{pmatrix}. \quad (12)$$

It is entrywise positive, has determinant one, and, at the base state, satisfies

$$B_{\gamma_*} \Omega_* B_{\gamma_*}^T = \Omega_*, \quad B_{\gamma_*}^T J_* B_{\gamma_*} = J_*.$$

This project-chosen word is a deterministic AGY-admissible existence witness; none of the operator theorems depends on choosing it from a finite periodic scan.

3 An all-length Rauzy matrix decoder

The main operator proof can isolate a branch by support and therefore does not require matrix injectivity. The next theorem nevertheless closes a second proposed alternative: different chronological branches cannot project to the same full matrix and cancel through opposite metaplectic signs in the present full-rank four-letter model.

The geometric antecedent is the standard simplicial-cylinder coding of Rauzy induction: finite words determine nested projective cones, with the two first-level cone interiors separated by the comparison hyperplane [5]. We do not claim novelty for that coding geometry. The statement below is a self-contained algorithmic form of it, tailored to the later-on-the-left matrix convention: a single cumulative matrix is inverted by a deterministic row-dominance subtraction.

Realization	Source status	C25 conclusion	Unresolved boundary
Scalar AGY C_b^1	Published spectral space	Not challenged	Scalar determinant questions are separate
Twisted $C_b^1(\mathcal{F})$	Literal project extension	Bounded, noncompact, nonnuclear	Holomorphic completion not inherited
Normalized $L^2(\mu; \mathcal{F})$	Natural Hilbert companion after section freeze	Contractive, noncompact; coisometry on $s = it$	Already a scalar L^2 space obstruction
Quasi-Hölder / holomorphic	Different completion	Not used for the no-go theorem	Branch localizers require a new proof

Table 1: The conclusion is tied to declared operator spaces. The raw C^1 obstruction detects the unsmoothed fibre; the normalized L^2 obstruction is already present for the scalar Perron–Frobenius operator.

Theorem 3.1 (Fixed-start path decoding). *Fix an irreducible labeled Rauzy permutation π . For every finite path γ starting at π , the pair (π, B_γ) , with the conventions (1)–(2), uniquely determines every directed arrow of γ .*

Proof. Let $\gamma = e_1 \cdots e_n$ and put $R = B_\gamma^T$. If the first edge has winner w and loser ℓ , then

$$R = (I + E_{w,\ell})R', \quad R' = B_{e_2 \cdots e_n}^T. \quad (13)$$

Consequently

$$\text{row}_w(R) = \text{row}_w(R') + \text{row}_\ell(R'), \quad \text{row}_\ell(R) = \text{row}_\ell(R'). \quad (14)$$

Every path matrix is nonnegative and unimodular. The true first edge therefore satisfies componentwise winner-row dominance

$$\text{row}_w(R) \geq \text{row}_\ell(R), \quad (15)$$

with at least one strict coordinate.

At a fixed irreducible state, the two outgoing arrows exchange the same rightmost top and bottom labels, so the other candidate requires the reverse dominance. If both candidates passed, the two rows would be equal. That is impossible because R is invertible. Thus exactly one candidate identifies the true first edge.

Peeling that edge is exact:

$$R' = (I - E_{w,\ell})R. \quad (16)$$

The operation replaces the winner row by winner minus loser, leaves a nonnegative integral matrix, and decreases the sum of all entries by the strictly positive loser-row sum. We update the labeled permutation and repeat. The entry sum cannot decrease indefinitely, and a valid path ends at the identity. Induction recovers the unique full word. $\square$

The proof yields a reproducible decoder.

Decoder. Start with the labeled state π and $R = B_\gamma^T$. Until $R = I$: (i) test the two outgoing winner/loser row dominances; (ii) select the unique passing edge; (iii) replace its winner row by winner minus loser; (iv) advance the labeled state. Reject an input matrix if neither or both candidates pass, the subtraction becomes negative, or the terminal matrix is not I .

Corollary 3.2 (Return-matrix freeness). *Return paths based at a fixed labeled state have distinct full Rauzy matrices. Concatenations of first-return paths form a free monoid on the countable first-return generators.*

Proof. Theorem 3.1 recovers the complete edge word. Visits to the fixed state then give its unique decomposition into first returns. $\square$

The word “full” matters. A characteristic polynomial, spectrum, or conjugacy class does not carry the row data required by (15). In a general stratum, restriction from relative to absolute homology can also discard information. For the present four-letter $\mathcal{H}(2)$ model, $\det \Omega_\pi = 1$, so no relative kernel is present. Matrix decoding recovers the path; the central metaplectic sign is then recovered by multiplying the declared edge lifts. We do not claim that an endpoint matrix without this lift convention already contains that sign.

The decoder also explains why retaining genuine chronology is practical. The ordered integer product is not merely a coarse endpoint statistic in this fixed-start coding: it admits a deterministic inverse. Replacing it by an averaged transition matrix, a spectrum, or a Parikh count destroys that inverse.

4 The raw vector-valued C_b^1 obstruction

Let

$$\mathcal{B}_{\mathcal{F}} = C_b^1(\Delta; \mathcal{F}), \quad \|F\|_{\mathcal{B}} = \sup_x \|F(x)\|_{\mathcal{F}} + \sup_x \|DF(x)\|.$$

The direct unsmoothed extension of (10) is

$$(\mathcal{L}_s F)(x) = \sum_{\gamma} w_{s,\gamma}(x) U_{\gamma} F(h_{\gamma} x), \quad w_{s,\gamma} = e^{-sr_{\gamma}} j_{\gamma}. \quad (17)$$

We first prove that this is a genuine bounded branch sum, not a formal expression.

Proposition 4.1 (Absolute branch convergence). *For every fixed s with $\Re s > -\sigma_0$, the series in (17) converges absolutely in $\mathcal{B}(\mathcal{B}_{\mathcal{F}})$.*

Proof. Set $a = \Re s$ and $v_{a,\gamma} = e^{-ar_{\gamma}} j_{\gamma}$. By (6), the logarithmic derivative of this positive weight is bounded by $C_J + |a|C_r$, uniformly in γ . Since Δ has finite diameter in the AGY Finsler metric, bounded distortion compares its supremum with its mean by a branch-independent constant. Change variables with h_{γ} and use the branch partition:

$$\sum_{\gamma} \|v_{a,\gamma}\|_{\infty} \leq C_a \sum_{\gamma} \int_{\Delta} e^{-ar_{\gamma}(x)} j_{\gamma}(x) dm(x) \quad (18)$$

$$= C_a \int_{\Delta} e^{-ar(y)} dm(y) < \infty. \quad (19)$$

The last inequality follows from (8) and $a > -\sigma_0$.

Differentiation gives

$$Dw_{s,\gamma} = w_{s,\gamma}(D \log j_{\gamma} - sDr_{\gamma}).$$

The product rule, $\|Dh_{\gamma}\| \leq \kappa$, and unitarity of U_{γ} imply

$$\|w_{s,\gamma} U_{\gamma}(\cdot \circ h_{\gamma})\|_{\mathcal{B} \rightarrow \mathcal{B}} \leq C(1 + |s|) \|v_{a,\gamma}\|_{\infty}.$$

Summing with (19) proves the claim. The same estimates hold for Bochner-valued derivatives because each U_{γ} is constant in the base coordinate. $\square$

The next compression is the decisive step. It uses a smooth localizer, not the discontinuous indicator of a branch cylinder.

Theorem 4.2 (Exact branch compression). *For every s with $\Re s > -\sigma_0$, the operator $\mathcal{L}_s$ in (17) is noncompact on $\mathcal{B}_{\mathcal{F}}$. In particular it is not a Banach nuclear operator.*

Proof. Fix a branch γ_0 and $x_0 \in \Delta$. Its preimage $y_0 = h_{\gamma_0}x_0$ lies in the open set D_{γ_0} . AGY's bump lemma supplies $\psi \in C_c^1(D_{\gamma_0})$ with $\psi(y_0) = 1$. Define bounded maps

$$J_\psi : \mathcal{F} \rightarrow \mathcal{B}_{\mathcal{F}}, \quad J_\psi v = \psi v, \quad R_{x_0} : \mathcal{B}_{\mathcal{F}} \rightarrow \mathcal{F}, \quad R_{x_0} F = F(x_0).$$

For $\gamma \neq \gamma_0$, the point $h_\gamma x_0$ belongs to the disjoint branch image D_γ and is outside the support of ψ . Absolute convergence from Proposition 4.1 permits evaluation of the full sum and gives the exact identity

$$R_{x_0} \mathcal{L}_s J_\psi = e^{-sr_{\gamma_0}(x_0)} j_{\gamma_0}(x_0) U_{\gamma_0}. \quad (20)$$

The scalar is nonzero. A nonzero multiple of a unitary on the infinite-dimensional space $\mathcal{F}$ is not compact. If $\mathcal{L}_s$ were compact, its bounded compression (20) would be compact, a contradiction. Nuclear operators between Banach spaces are compact, so $\mathcal{L}_s$ is not nuclear. $\square$

The proof also gives a quantitative lower bound

$$\|\mathcal{L}_s\|_{\text{ess}} \geq \frac{|e^{-sr_{\gamma_0}(x_0)} j_{\gamma_0}(x_0)|}{\|R_{x_0}\| \|J_\psi\|} > 0. \quad (21)$$

No conclusion is drawn from a single noncompact summand. Equation (20) isolates one branch in the already convergent full operator, so cancellation elsewhere is impossible.

5 The normalized Hilbert companion

The invariant density in (9) defines conditional inverse-branch probabilities

$$p_\gamma(x) = j_\gamma(x) \frac{\rho(h_\gamma x)}{\rho(x)}, \quad \sum_\gamma p_\gamma(x) = 1 \quad \text{a.e.} \quad (22)$$

Let $\mathcal{H} = L^2(\Delta, \mu; \mathcal{F})$ and define

$$(\mathcal{P}_s F)(x) = \sum_\gamma p_\gamma(x) e^{-sr_\gamma(x)} U_\gamma F(h_\gamma x). \quad (23)$$

This Hilbert companion is not AGY's claimed scalar spectral-gap space. It is a natural invariant-measure normalization used here as an independent robustness test.

The countable Bochner sum in (23) is well defined. Indeed, for $a = \Re s \geq 0$, Cauchy–Schwarz gives

$$\sum_\gamma p_\gamma e^{-ar_\gamma} \|F(h_\gamma x)\| \leq \left(\sum_\gamma p_\gamma \right)^{1/2} \left(\sum_\gamma p_\gamma \|F(h_\gamma x)\|^2 \right)^{1/2}. \quad (24)$$

The square of the second factor has integral $\|F\|_{\mathcal{H}}^2$ by branch disintegration, so the sum is absolutely convergent almost everywhere. For any tail A of an enumeration, the same inequality yields

$$\left\| \sum_{\gamma \in A} p_\gamma e^{-sr_\gamma} U_\gamma F \circ h_\gamma \right\|_{\mathcal{H}}^2 \leq \int_{\bigcup_{\gamma \in A} D_\gamma} \|F(y)\|^2 d\mu(y),$$

which tends to zero. Hence the series converges unconditionally in $\mathcal{H}$, independently of its enumeration.

Theorem 5.1 (Contraction and branch norm). *For $\Re s \geq 0$, $\mathcal{P}_s$ is a contraction on $\mathcal{H}$. For one branch, let*

$$(K_{s,\gamma} f)(x) = p_\gamma(x) e^{-sr_\gamma(x)} f(h_\gamma x), \quad K_{s,\gamma} : L^2(D_\gamma, \mu) \rightarrow L^2(\Delta, \mu).$$

Then

$$\|K_{s,\gamma}\| = \operatorname{ess\,sup}_{x \in \Delta} \sqrt{p_\gamma(x)} e^{-\Re(s)r_\gamma(x)} > 0. \quad (25)$$

Because μ is nonatomic,

$$\|K_{s,\gamma}\|_{\operatorname{ess}} = \|K_{s,\gamma}\|. \quad (26)$$

Moreover,

$$\|\mathcal{P}_s\|_{\operatorname{ess}} \geq \|K_{s,\gamma}\| > 0, \quad (27)$$

so $\mathcal{P}_s$ is noncompact.

Proof. Convexity of the squared Hilbert norm and (22) give, for $a = \Re s \geq 0$,

$$\|\mathcal{P}_s F(x)\|^2 \leq \sum_{\gamma} p_\gamma(x) e^{-2ar_\gamma(x)} \|F(h_\gamma x)\|^2 \leq \sum_{\gamma} p_\gamma(x) \|F(h_\gamma x)\|^2.$$

The branch disintegration identity

$$\int_{D_\gamma} q(y) d\mu(y) = \int_{\Delta} p_\gamma(x) q(h_\gamma x) d\mu(x) \quad (28)$$

then proves $\|\mathcal{P}_s F\| \leq \|F\|$.

Applying (28) to one scalar branch yields

$$\|K_{s,\gamma} f\|^2 = \int_{D_\gamma} p_\gamma(Ty) e^{-2ar_\gamma(y)} |f(y)|^2 d\mu(y),$$

which gives (25). Extension by zero from $L^2(D_\gamma, \mu)$ isolates this branch even for the scalar normalized operator.

To prove (26), write

$$m(y) = \sqrt{p_\gamma(Ty)} e^{-ar_\gamma(y)}, \quad M = \operatorname{ess\,sup}_{D_\gamma} m = \|K_{s,\gamma}\|.$$

For every $\varepsilon > 0$, the set $\{m > M - \varepsilon\}$ has positive measure. Nonatomicity lets us split it into pairwise disjoint positive-measure sets E_n . The normalized indicators $f_n = 1_{E_n} / \sqrt{\mu(E_n)}$ are orthonormal, hence weakly null, while $\|K_{s,\gamma} f_n\| \geq M - \varepsilon$. Every compact operator sends f_n to zero in norm. Thus $\|K_{s,\gamma}\|_{\operatorname{ess}} \geq M - \varepsilon$; let $\varepsilon \downarrow 0$. The reverse inequality is automatic.

Extension by zero from $L^2(D_\gamma, \mu; \mathcal{F})$ isolates the branch exactly:

$$\mathcal{P}_s J_\gamma = K_{s,\gamma} \otimes U_\gamma. \quad (29)$$

For a bounded scalar operator K and an infinite-dimensional unitary U , $\|K \otimes U\|_{\operatorname{ess}} = \|K\|$. Applying this identity to (29) proves (27). With scalar fibre $U_\gamma = 1$, the same compression and (26) prove noncompactness throughout $\Re s \geq 0$ as well. $\square$

On the imaginary axis the obstruction is sharp.

Theorem 5.2 (Coisometry). *For every $t \in \mathbb{R}$,*

$$\mathcal{P}_{it} \mathcal{P}_{it}^* = I, \quad \|\mathcal{P}_{it}\|_{\operatorname{ess}} = 1. \quad (30)$$

Proof. Take inner products linear in the first argument. On $y \in D_\gamma$, define

$$(\mathcal{U}_t G)(y) = e^{itr_\gamma(y)} U_\gamma^* G(Ty).$$

Invariance of μ and unitarity make $\mathcal{U}_t$ an isometry. Equations (23) and (28) give $\mathcal{P}_{it} = \mathcal{U}_t^*$, hence the first identity in (30). In the Calkin algebra, $[\mathcal{P}_{it}][\mathcal{P}_{it}]^* = [I]$, so $\|\mathcal{P}_{it}\|_{\operatorname{ess}} \geq 1$. Contractivity gives the reverse inequality. $\square$

The normalized L^2 obstruction is not caused specifically by the oscillator fibre: the scalar operator is noncompact throughout $\Re s \geq 0$, and the Koopman-adjoint argument makes it a coisometry with essential norm one on the imaginary axis. This section therefore closes a proposed Hilbert realization as a space-level robustness test; the oscillator-specific multi-branch obstruction is Theorem 4.2.

5.1 The absolute atomic half-plane

Let $q_\gamma = \mu(D_\gamma)$ and $R_\gamma = \inf r_\gamma$. Uniform distortion, (7), and (9) imply

$$p_\gamma(x) \asymp e^{-dr_\gamma(x)}, \quad q_\gamma \asymp e^{-dR_\gamma}. \quad (31)$$

Thus the branch operator extended by zero satisfies

$$\|K_{s,\gamma}\| \leq C q_\gamma^{1/2 + \Re(s)/d}. \quad (32)$$

When $\Re s \geq d/2$, the exponent is at least one and $\sum_\gamma q_\gamma = 1$, so

$$\sum_\gamma \|K_{s,\gamma}\| < \infty. \quad (33)$$

For four labels the threshold is $\Re s \geq 2$.

Even if branch terms are grouped by their projected symplectic matrix, no signed aggregate can vanish. Choose an input supported in one D_{γ_0} . Every other branch evaluates that input outside its support, while the γ_0 term remains nonzero. This proof retains the central sign rather than replacing it by an absolute value. The decoder in Section 3 independently shows that distinct fixed-start branches have distinct full matrices in the current full-rank model.

6 Exact certificate and independent checks

The functional-analytic obstruction is a theorem and does not rest on a finite numerical pattern. Exact computation has two narrower roles: it specifies one project-chosen AGY-admissible section in the selected four-letter class and tests the decoder implementation against mutations.

The producer starts from (1234)/(4321), reconstructs all seven labeled states and fourteen arrows, and verifies (3) on every edge. It then follows every one of the 128 arrows of (11), records winner and loser labels, builds (12) with later edges on the left, and checks the eight complete blocks. It computes the maximal initial constant-type run rather than trusting the compressed notation: the run has length 65 because η begins with t .

For a rational interior witness, the code puts

$$x_0 = \frac{B_{\gamma_*}^T \mathbf{1}}{\mathbf{1}^T B_{\gamma_*}^T \mathbf{1}}, \quad y_0 = \frac{B_{\gamma_*}^T x_0}{\mathbf{1}^T B_{\gamma_*}^T x_0}. \quad (34)$$

It differentiates the projective map in three independent simplex coordinates and verifies exactly

$$\det Dh_{\gamma_*}(x_0) = (\mathbf{1}^T B_{\gamma_*}^T x_0)^{-4} = e^{-4r_{\gamma_*}(x_0)}. \quad (35)$$

The independent checker does not import the producer. It has separate implementations of the Rauzy move, crossing form, matrix multiplication, spanning-tree frames, fixed-fibre edge matrices, row decoder, projective derivative, and first-return enumeration. The recorded checks are summarized in Table 2.

All eleven independent validation gates and all fourteen registered regression/mutation tests pass. No numerical metaplectic central sign is selected: the frame certificate proves that edge lifts can be chosen and composed coherently, while both possible signs remain in scope.

Object	Exact result	Logical role
Rauzy graph	7 states, 14 transported edges	Source lock
State trivialization	7 integral frames; 14 fixed-fibre edges (6 identity, 8 nonidentity)	Coherent pathwise lift
Section word	length 128, 8 complete blocks	Strong positivity
Neatness	initial run 65; no proper border	Return-model lock
Section matrix	positive, determinant 1, symplectic	Fibre applicability
Decoder replay	all 128 section arrows recovered	Literal chronology check
First-return stress	35,420 words through length 22	Non-proof regression sentinel
Matrix collisions in stress window	0	Diagnostic only
Projective derivative	exponent exactly 4	Jacobian convention lock

Table 2: Exact certificate claims. The all-length no-collision statement is Theorem 3.1; the finite stress window is not substituted for that proof.

The stress language is based at the central seed

$$\pi_c = (1234)/(4321),$$

which is distinct from the AGY section base $\pi_* = (1342)/(4321)$. State elimination at π_c gives the central first-return generating function

$$F(z) = \frac{2z^3}{1 - z - z^2}, \quad (36)$$

so the number of first returns of elementary length $n \geq 3$ is $2F_{n-2}$, with $F_1 = F_2 = 1$. This exact rational language is useful for checking enumeration completeness. It is not an arithmetic prime law and does not improve the Route-A orbit verdict.

The checker also registers negative tests for transposed chronology, winner/loser reversal, the wrong Jacobian exponent, a mutated section word, an incorrect initial-run count, lost labeled state, and confusion between a short central return and the AGY γ_* -section. All released equalities use integers or rationals; no floating tolerance is needed.

7 Route-A decision, limitations, and conclusion

The ordinary-determinant proposal fails on both declared spaces. On the raw Banach space, Theorem 4.2 excludes Banach nuclearity for every $\Re s > -\sigma_0$. On the normalized Hilbert space, Theorem 5.1 excludes compactness and every finite Schatten class for $\Re s \geq 0$; consequently neither the ordinary Fredholm determinant nor a finite-order regularized Schatten determinant is available. Theorem 5.2 makes the obstruction quantitative on the imaginary axis.

Noncompactness alone does not imply that $I - z\mathcal{P}_s$ is never Fredholm or invertible. It also does not rule out a determinant defined by a flat or dynamical trace, a Weil distribution character on its regular locus, a semifinite trace forced by geometry, or genuine continuous group smoothing. These constructions change the trace framework and require their own well-posedness theorems.

A holomorphic or anisotropic completion is the most precise escape exposed by the proof. A nonzero holomorphic function cannot have compact support in one real branch, so the bump/evaluation compression does not transfer to that setting. Conversely, scalar compactness on a holomorphic space does not automatically persist after tensoring with an unsmoothed infinite-dimensional unitary fibre. A viable continuation must fix a common complex domain, prove complex distortion and branch-norm estimates, retain the chronological central sign, and establish a legal trace or determinant.

The quasi-Hölder space of Aimino et al. [1] illustrates the same need for space-specific care. It contains characteristic functions of some sets and is a Banach algebra, but the cited paper does not prove that every return cylinder has the boundary regularity needed for its indicator to belong to the space. We therefore leave that application conditional rather than silently importing an L^2 projection.

Under the project’s Route-A rubric, the result is

(A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT).

The all-length decoder provides a natural reproducible orbit coding, but no prime-like clock or arithmetic amplitude law. The tested ordinary dynamical determinant is obstructed before a zero comparison is meaningful. There is no Riemann divisor, functional equation, gamma factor, continuation of the metaplectic determinant, or Riemann–von Mangoldt law. The symplectic cocycle supplies a natural oscillator fibre, but no surviving self-adjoint Hilbert–Pólya operator. Route B is therefore not authorized.

The appropriate next step is a system-space switch: either source-lock a holomorphic no-localizer transfer family or construct a geometrically forced generalized trace. Extending the elementary periodic ledger beyond the existing cutoff would not address the failed operator gate.

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A Auxiliary operator facts and reproducibility details

Lemma A.1 (Tensor essential norm). *Let $K : H_0 \rightarrow H_1$ be bounded between Hilbert spaces and let U be unitary on an infinite-dimensional Hilbert space F . Then*

$$\|K \otimes U\|_{\text{ess}} = \|K\|.$$

Proof. For a unit vector $x \in H_0$ and an orthonormal sequence (e_n) in F , the vectors $x \otimes e_n$ converge weakly to zero. Every compact operator C sends them to zero in norm, whereas

$$\|(K \otimes U)(x \otimes e_n)\| = \|Kx\|.$$

Thus $\|K \otimes U - C\| \geq \|Kx\|$. Taking the supremum over x proves the lower bound; the operator norm gives the reverse bound. $\square$

A.1 Why the branch norm contains a square root

The normalized scalar branch is $K_{s,\gamma}f = p_\gamma e^{-sr_\gamma} f \circ h_\gamma$. Applying (28) once removes one of the two factors of p_γ :

$$\|K_{s,\gamma}f\|^2 = \int_{D_\gamma} p_\gamma(Ty) e^{-2\Re(s)r(y)} |f(y)|^2 d\mu(y).$$

Hence the norm is the essential supremum of $\sqrt{p_\gamma(Ty)} e^{-\Re(s)r(y)}$, not of $p_\gamma(Ty) e^{-\Re(s)r(y)}$. This factor is responsible for the absolute-atom threshold $\Re s \geq d/2$.

A.2 Decoder invariants

At every decoder step, the two outgoing edges exchange winner and loser. For a valid image matrix, exactly one row difference is componentwise nonnegative. The checker records both differences, the selected edge, the matrix before and after subtraction, and

$$\sum_{i,j} R_{ij} - \sum_{i,j} R'_{ij} = \sum_j R_{\ell j} > 0.$$

This trace detects three common bugs independently: using $B_1 \cdots B_n$ instead of later-on-left chronology, peeling columns instead of rows, and advancing the state before selecting the edge.

A.3 Reproduction contract

The producer and checker are separate Python programs. The checker imports neither producer functions nor producer constants. Both reconstruct the graph from the literal seed, while their move, multiplication, crossing-form, decoder, and projective-derivative routines are separately written. The certificate stores a Material Passport, runtime versions, source hashes, and exact JSON values. The repository runner executes the producer, checker, unit tests, mutation tests, and artifact-hash generation and verification. The paper has a separate clean-build command and compilation report.